

Optical SLAVE FLASH TRIGGER

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In digital cameras, the slave flash is part of an external flashgun (or speedlight) that sits quiet until it sees a bright light. When the slave flash sees a light from another source, it flashes to trigger the speedlight, which gives a more powerful lighting option than a typical built-in flash unit.

Use of off-camera flashgun option gives you some great dynamic lighting for wonderful photography that would be difficult otherwise. Here we describe the circuit of a light-sensitive sensor that detects light impulses and wakes up the electronics to fire the attached speedlight. The goal is to make the circuit small, cheap, isolated and independently powered.

The circuit uses inexpensive and readily available components. Its main functions are:

1. Sense the light impulse (from

the master flash) using a standard light sensor

2. Verify that the light level is above a constant threshold (which can be altered using a potentiometer to tweak the detection sensitivity)

3. Trigger the slave flash through a galvanically-isolated (contact-free) electronic switch

The author's prototype is shown in Fig. 1.

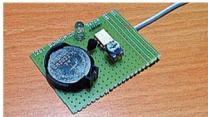


Fig. 1: Author's prototype without enclosure

Circuit and working

Fig. 2 shows the circuit of optical flash trigger. It is built around phototransistor L14F1 (T1), opto-isolated triac MOC3021 (IC1), npn transistor BC547 (T2) and a few other components.

This circuit works on the principle of Wheatstone bridge. When light falls on the phototransistor (T1), its conductivity (and current) increases, which forward biases the driver transistor (T2). As a result, IC1 wakes up to trigger the slave flash; an optical coupling makes it safe to use even the oldest flashgun that could otherwise damage your circuitry by high-voltage ignition.

However, note that the connected slave flash doesn't work in bright daylight or bright ambient light. Likewise, it may even misfire when you have a very bright source of artificial light in its path. The potmeter (VR1) helps you to set correctly the light detection threshold level.

The light sensor used here is a generic 5mm round phototransistor L14F1

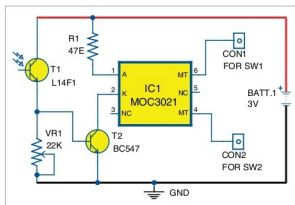


Fig. 2: Circuit of optical slave flash trigger

PARTS LIST

Semiconductors:

- IC1 - MOC3021 opto-isolated triac
- T1 - 5mm round L14F1 phototransistor

- T2 - BC547 npn transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1 - 47-ohm
- VR1 - 22-kilo-ohm potmeter

Miscellaneous:

- CON1, CON2 - 1-pin connector
- BATT.1 - CR2032, 3V coin cell (Li-MnO₂)

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